

Classifying life

Scientists have described more than a million and a half different species of living things. They classify them into groups based on how they are related.

There are lots of ways of classifying organisms. Insects, birds, and bats could be grouped as flying animals, and plants could be grouped by how we use them. But neither of these systems shows natural relationships. Biological classification works by grouping related species. Bats, for instance, have closer links to monkeys than they do to birds, because they are both furry mammals that have evolved from the same mammal ancestors.

Single-celled organisms are the most common form of life in some kingdoms—including archaea, bacteria, and protozoans.

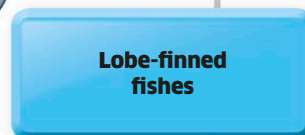
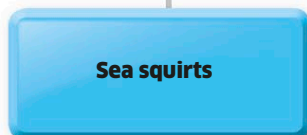
Seven kingdoms of organisms



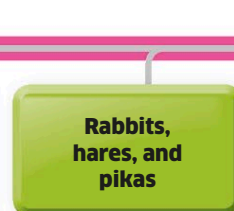
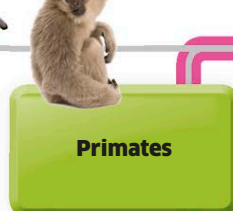
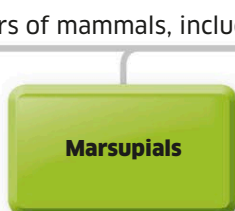
More than 30 phyla of animals, including ...



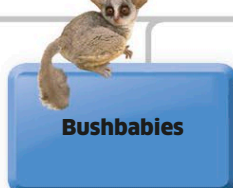
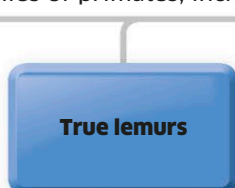
12 classes of chordates, including ...



29 orders of mammals, including ...



15 families of primates, including ...



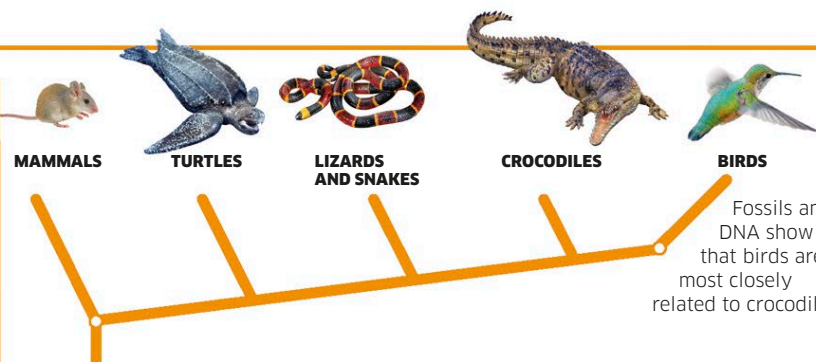
Scientific names

Every species has a two-part scientific name using Latin words that are internationally recognized in science. The first part identifies its genus group, the second its species. Lions and tigers belong to the *Panthera* genus of big cats, but have different species names.



Panthera leo
Lion

Panthera tigris
Tiger



Fossils and DNA show that birds are most closely related to crocodiles.

Classifying birds

Modern classification aims to show how organisms are linked by evolution. Birds and reptiles are traditionally in two separate classes. But birds evolved from reptile ancestors (see pp.136–137), so many scientists think they should be a sub-group of the reptiles.